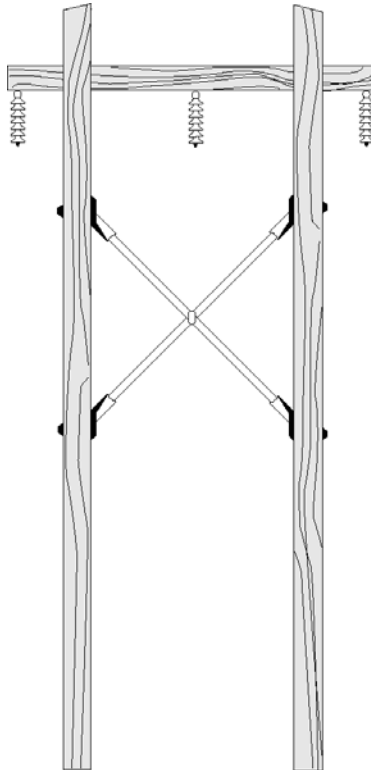




Western Area
Power Administration

TECHNICAL SPECIFICATION CROSSARMS AND BRACES



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SAFETY
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CROSSARMS AND BRACES

SECTION 1-- CROSSARMS AND BRACES

1.1 GENERAL:

Furnish and deliver crossarms and braces as stated in these technical specifications. Materials shall be of the type and grade most suitable for the application and shall conform to industry standards and practices. All materials shall be free from defects, of recent manufacture, and unused.

Referenced Documents:

- American National Standards Institute (ANSI) Standards latest edition, Specifications and Dimensions for solid sawn-wood crossarms and braces (ANSI 05.3) and structural glued laminated timber for utility structures (ANSI 05.2) Standards latest editions
- American Institute of Timber Construction (AITC), AITC 117 and ANSI/AITC A190.1 Standards latest editions
- American Wood Preservers Association (AWPA) Standards latest editions
- West Coast Timber Inspection Bureau (WCLIB) Standards latest editions

1.2 MATERIAL:

1. CROSSARMS:

(1) Glued Laminated:

- 1) General: The laminating stock shall consist of Pacific Coast Douglas Fir.
- 2) Manufacture: Material, manufacture, and quality control shall be in conformance with ANSI/AITC A190.1, ANSI 05.2, and AITC 117.
- 3) Laminating Combinations: Shall meet the requirements of ANSI/AITC A190.1, Structural Glued Laminated Timber, shall have a balanced lay-up meeting 24F-V8, with tension laminations conforming to AITC 302-20, free of heart centers, and provide a minimum ultimate fiber strength of 6800 psi.
- 4) Adhesives: Wet-use type complying with ANSI/AITC A190.1 and ASTM D 2559. Melamine urea adhesives shall not be used.
- 5) Appearance: As specified in AITC 110, Industrial Appearance Grade, and ANSI 05.2. Edges of crossarms shall be eased to 3/16-inch radius.

(2) Drilling: Crossarms shall be drilled before treatment in accordance with a Western-approved sketch showing crossarm drilling details. Holes in crossarms shall be drilled 1/8 inch larger than their respective bolt diameters.

(3) Cross Section: Crossarms shall have a finished cross section as shown on the drawings.

2. WOOD BRACES:

(1) Solid-Sawn:

- 1) Lumber: Seasoned Pacific Coast Douglas Fir, cut from live timber, and kiln-dried in accordance with AWPA C25.

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- 2) Manufacture: Manufactured in accordance with ANSI 05.3, free of heart center (FOHC), surfaced four sides (S4s) after seasoning, and incised four sides to depth of 3/16 inch. Lumber shall provide an ultimate design value of 6,600 psi in bending.

Maximum knot sizes for braces shall be as follows:

Face Width (inches)	Knot Size (inches)
3 3/8 and 3 5/8	3/4
4 3/8	1
5 3/8	1 1/8
8 1/2	1 1/2

- (2) Glued Laminated:

- 1) General: The laminating stock shall consist of Pacific Coast Douglas Fir.
 - 2) Manufacture: Material, manufacture, and quality control shall be in conformance with ANSI/AITC A190.1 and AITC 117.
 - 3) Laminating Combinations: Shall meet the requirements of ANSI/AITC A190.1, Structural Glued Laminated Timber and shall be fabricated using lumber meeting the requirements for combination 5 as shown in Table B2 of the AITC 117.
 - 4) Adhesives: Wet-use type complying with ANSI/AITC A190.1 and ASTM D 2559. Melamine urea adhesives shall not be used.
 - 5) Appearance: As specified in AITC 110, Industrial Appearance Grade and ANSI 05.2.
- (3) Drilling: Braces shall be drilled before treatment. Holes in braces shall be drilled 1/8 inch larger than their respective bolt diameters.
- (4) Cross Section: Wood braces shall have a finished cross section as shown on the drawings.

3. KNEE BRACES:

- (1) Fiberglass: Outside knee braces shall be made of fiberglass with a minimum ultimate strength of 25,000 pounds

1.3 QUALITY CONTROL:

1. Glued Laminated Timber: In accordance with ANSI/AITC A190.1.
2. Solid-Sawn Timber: In accordance with paragraph 169a, "Select Structural Transmission Crossarms, Planks, and Timbers", of the West Coast Lumber Inspection Bureau Grading Rules No. 17. Moisture content shall not exceed 22 percent after seasoning, and the moisture content gradient shall not exceed 5 percent between the center and surface of the crossarm or brace.

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1.4 WOOD TREATMENT:

1. Glued laminated timber shall be incised on all four sides and treated with a preservative in accordance with AWP A T1 or as stated herein.
2. Solid-sawn timber shall be incised on all four sides and treated with a preservative in accordance with AWP A T1 or as stated herein.
3. Pentachlorophenol and copper naphthenate shall meet the requirements of AWP A P8. Solvent for pentachlorophenol and copper naphthenate shall be Type A per AWP A P9. The minimum retention of copper naphthenate preservative shall be:

Thickness of Crossarm (inches)	Assay Zone (inches from surface)	Copper Naphthenate (pounds/cubic foot as copper metal)
under 5	0.4	0.040
over 5	0.5	0.040

Penetration to 2 1/2 inches is required longitudinally from bolt holes and ends.

4. Timber shall be clean and dry to the touch after treatment.

1.5 MARKING:

Glued laminated timber shall be marked in accordance with ANSI 05.2. Solid sawn timber shall be marked in accordance with ANSI 05.3.

1.6 CERTIFICATIONS:

As soon as practical after completion of treatment, provide one certified copy of the records or test results to the COR:

1. TREATMENT TESTS: Crossarms and braces shall be inspected in accordance with AWP A M2 to assure compliance with the requirements in AWP A, or as specified herein. The penetration of preservative shall be tested in accordance with the requirements of AWP A A3. The retention of preservative shall be tested in accordance with the requirements of AWP A A5 or AWP A A9.

Treatment test results shall show:

- (1) Compliance with applicable AWP A specifications.
 - (2) Number and type of pieces per charge.
 - (3) Specie of wood.
 - (4) Kind of preservative used.
 - (5) Penetration obtained.
 - (6) Retention obtained.
 - (7) Moisture content after treatment.
2. PRESERVATIVE TESTS: The preservative used in the treating of crossarms and braces shall be tested in accordance with AWP A M2 and AWP A A5 (Method 5 for pentachlorophenol and Method 7 for copper naphthenate).

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1.7 INSPECTIONS:

1. All work performed shall be subject to inspection, in accordance with AWP A Standard M2 and ANSI/AITC A190.1 and AITC 200, and no crossarms shall be shipped until all required inspections have been made. The Contractor shall furnish reasonable facilities, including skids, blocks, sufficient space and personnel to provide for laying out in an orderly manner and handling and turning a reasonable number of crossarms at a time for inspection. Inspection will be made before and after treatment unless one or both inspections are waived. The Government inspector shall have the right to take samples of the preservative solution or ingredients thereof, whenever he may elect to do so. Such samples shall be analyzed whenever the Government may desire to have the analysis made. Acceptance of the materials or the waiving of the inspection thereof shall in no way relieve the Contractor of the responsibility for furnishing materials meeting the requirements of this specification.

1.8 STORAGE, HANDLING, AND TRANSPORT:

Crossarms and braces shall be protected from elements and stored off ground.